

Nandani Gupta

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EDUCATION

Indian Institute of Technology Jammu, India
Bachelor of Technology in Chemical Engineering

2021 – 2025

WORK EXPERIENCE

ML / Backend Engineer | Lohum Cleantech Pvt. Ltd. (Greater Noida, India)

Tech Stack: Python, LangChain, RAG, Scikit-learn, Vector DB, FastAPI, PostgreSQL, AsyncIO, Docker

July 2025 – Present

- Built a **RAG-based document intelligence system** using **LangChain** and a **vector database** – indexes **500+ regulatory documents** via semantic embeddings, retrieves context-relevant chunks at query time, and surfaces structured insights through a **FastAPI** service; reduced manual research time for operational teams measurably in production.
- Engineered an **AsyncIO** data ingestion pipeline polling **PIB and email sources 24×/day** – performs **document diffing** to detect changes, triggers selective re-chunking and **embedding generation** only for updated content, and persists structured metadata to **PostgreSQL** for downstream retrieval and analytics.
- Served **ML model inference** through **REST API** endpoints, wrote unit tests across all pipeline modules, containerized services with **Docker**, and iterated on solutions with cross-functional stakeholders within agile sprint cycles.

Software Developer Intern | Arvon Web Solutions (Remote)

Tech Stack: Python, Node.js, C++, PostgreSQL, REST APIs, Docker, Git

May 2024 – July 2024

- Built **RESTful backend services** in **Python** and **Node.js** for a SaaS hospital management platform – designed **normalised PostgreSQL schemas** for patient records and scheduling, wrote automated **data validation scripts**, and executed **unit and integration tests** across all endpoints before each sprint release.

INTERNATIONAL EXPOSURE

Research Intern | Hanyang University, South Korea

Tech Stack: Python, MATLAB, NumPy, SciPy, Data Analysis, Scripting

May 2024 – July 2024

- Built **Python** data processing and analysis pipelines within a cross-functional international research team – applied structured numerical analysis and **signal processing** techniques, iterated on calibration models, and communicated findings to senior stakeholders across disciplines.

PROJECTS

Fintech Fraud Risk Engine | Real-Time ML-Powered Risk Decision System

[GitHub](#)

Tech Stack: Python, Scikit-learn, FastAPI, Redis Streams, PostgreSQL, Pandas, Docker, SHAP

- Built an end-to-end **fraud detection platform** for financial transactions – trained a **Logistic Regression** classifier on real credit card transaction data achieving **ROC-AUC 0.98** with a full **feature scaling and inference pipeline**; fused ML fraud scores, **KYC risk profiling** (geography, credit score, tenure rules), and **velocity analytics** into a weighted risk score mapped to **APPROVE / REVIEW / DECLINE** decisions.
- Implemented **SHAP explainability** for every model inference, ingested live transactions via **Redis Streams** producer-consumer pipeline, persisted all decisions to a **PostgreSQL** audit ledger, and deployed the full stack as **Dockerized FastAPI microservices** – applying systematic **regression testing** on ML inference pipelines using **Pandas**.

Crude Oil Process Analytics | ML-Based Fault Detection & Yield Prediction

[GitHub](#)

Tech Stack: Python, Scikit-learn, SciPy, NumPy, Pandas, PCA, Matplotlib, Seaborn

- Built a **yield prediction model** using **Scikit-learn** regression on API gravity, vapour pressure, and distillation temperatures achieving **$R^2 = 0.88$** ; implemented **PCA-based multivariate fault detection** using **Hotelling's T^2** and **Q statistics** to identify deviations from normal refinery operating conditions – both modules validated against real process datasets with visualised outputs.
- Modelled **hydrocracking reaction kinetics** via **ODE systems** solved with **SciPy**, applied **Singular Value Decomposition (SVD)** to identify fast and slow reaction time-scale dynamics, and generated **concentration profiles and phase portraits** – demonstrating applied numerical ML and process analytics across a full data science pipeline.

Movie Recommender System | Content-Based ML Recommendation Engine

[GitHub](#)

Tech Stack: Python, Scikit-learn, Pandas, NumPy, CountVectorizer, Cosine Similarity

- Built a **content-based filtering recommendation engine** using **CountVectorizer** to extract feature vectors from movie metadata (genres, cast, crew, keywords) and **cosine similarity** to rank and retrieve the top-N most similar movies – demonstrating core **NLP feature engineering** and **similarity-based retrieval** techniques foundational to recommendation and ranking systems used in fintech personalisation.

TECHNICAL SKILLS

ML & Data Science: Scikit-learn, Pandas, NumPy, SciPy, PCA, Logistic Regression, SHAP, Feature Engineering, NLP (CountVectorizer, TF-IDF), Cosine Similarity, ODE Modelling, SVD

AI / LLMs: LangChain, RAG, Vector Databases, Embedding Generation, OpenAI API, Prompt Engineering

Backend & Infra: Python, FastAPI, Node.js, Golang, REST APIs, Redis (Streams + Cache), PostgreSQL, AsyncIO, Docker, Git

Core CS: Data Structures & Algorithms (450+ LeetCode), System Design, OOP, Distributed Systems, Event-Driven Architecture

PUBLICATIONS & ACHIEVEMENTS

- Solved **450+** DSA problems on **LeetCode** – strong analytical and problem-solving skills across algorithms, data structures, and mathematical reasoning.
- Published in **Springer SCOPUS**: “Numerical investigation of Spray characteristics of lean-burn gas turbine injector”, FMFP 2023, IIT Jodhpur. [Publication Link](#)